

Coursework 1

TEAM 15

Team Number: 15

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Links:

Miro:

https://miro.com/welcomeonboard/UzhiV2pNYIJZQIA1bk1CaU9NRGZNQ3RueIVCYUxjK1B3d2pyK01CUU96bEpSK2tSTzJydjQyalNjeJFQWE5BQWFhall5TkN4QUtDL2JYWXI3QUZZeEZyMTJpUHHJuM3FqRDcvNnhmQIVHSjJRMjh2Z0tpbm5KdU1vaHh2bnRCVnZ3VHhHVHd5UWtSM1BidUtUYmxycDRnPT0hdjE=?share_link_id=895122050498

VFM Miro:

https://miro.com/welcomeonboard/dWt5UXB3ZmpSMGRhVEFFaGNjOFVuUzlkWUFtN1ZOMExVZzRsWnVFU2xYT0FRYmJrTmFRdmhXeTdMMmlzRDBOYjBkTDduV1gyaTRzaU4vL3dWSG1UYIZyMTJpUHHJuM3FqRDcvNnhmQIVHSjNrUWljK2tZWktKYW5Uam84NytRWU1BS2NFMDfkcUNFSnM0d3FEN050ekl3PT0hdjE=?share_link_id=960710679362

Teams:

[Team-15 | COM435 \(2025-2026\) - Lab Group 1 | Microsoft Teams](#)

Ungroomed Product Backlog

Story ID	Feature title	User Story Narrative
S1	Elderly Mode	As an elderly person I would like a user interface that has large text, big buttons and clear icons
S2		As an elderly person I want clear confirmation of my purchases and any errors that may occur
S3	Accessibility - Audio/visual	As a blind user, I would need the app to have audio assistance so I can book.
S4		As a disabled user, I want a simple start/ stop listening control for voice commands
S5		As a disabled user I would like to search the app using only my voice
S6		As a daily user, I would like to be able to book using my voice to save time typing
S7		As a user, I would like the system to repeat if it doesn't understand my commands
S8		As a visually impaired user I want a high contrast colour scheme, so icons and texts are easier to see

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S9		As a user I want to switch between light and dark themes to help with reducing eye strain
S10		As a daily user, I want my contrast and font option saved so I don't need to keep enabling this
S11	Accessibility Friendly routes	As a wheelchair user I want journey options that will include wheelchair friendly stations
S12		As a disabled user I want to see an option for step free routes so I can get around easily
S13	Student Tariff Pricing	As a student who commutes to university, I want to be able to travel affordably. Having the opportunity to get a discounted travel with student pricing would release a lot of stress.
S14		As a student I would like a verified student account so that I can access discounts for my travel instantly without needing to upload proof every time
S15		As a student I want tickets to automatically display my discounted price
S16		As a student on a tight budget, I want to see the percentage and the amount of the discount
S17	Online Bus/Train Timetables	As a commuter I want to see real time bus departures for chosen stops so I can time my journey more accurately
S18		As an ordinary app user, I want to have access to bus /train timetable anywhere at any time so I could plan my route
S19		As a daily commuter who phone has poor battery, I would like the option to print the online timetable without needing to worry about my phone battery
S20		As a daily commuter, I would like it to automatically recommend my most common route, to save time entering details again
S21	In-App Ticket Purchases	As a commuter I want to be able to purchase my ticket easily and reliably, highlighting any issue immediately so that i can get to my volunteering work on time.

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S22		As a user I want to securely store a payment method so I can buy tickets quickly
S23		As a frequent Traveller, I want to top up my travel card to avoid queues
S24		As a student on a budget, I would like my in-app purchases sent to my email for 2nd way of confirmation
S25	Multi-Language support	As an international tourist, I want to be able to view information in my preferred language so I can book correctly
S26		As an international tourist, I want the app to detect my country and automatically put it in my language for me
S27	Currency Converter	As a person who is sightseeing in Dublin, I want to convert ticket prices within the app. To help me keep track of costs, so I stay within the budget and can enjoy my time travelling.
S28		As an international tourist I would like the tickets converted into my home currency so I can prepare my transport
S29		As a user, I want it to use my location and default to the currency in my location
S30		As a user, I want it to show me the exchange rate so I can be more informed about the costs

User Stories

Story ID: 1

Feature Title: An Accessible UI Consistency

Story Title: Elderly Mode

Story Narrative: As an elderly user, I want to be able to enable an Elderly Mode so that I can easily read text and navigate using larger buttons and icons due accessibility needs.

Story Size: Small

Acceptance Criteria Scenarios:

Scenario Title: Elderly Mode is initiated

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Given I have sight problems

When I turn on elderly Mode

Then the app then adjusts the UI to display larger text, buttons, and icons

Scenario Title: Inconsistent UI in elderly mode

Given I struggle to use complex UI and Elderly Mode is enabled

When I navigate through different pages within the app

Then the text and buttons on some pages are smaller than other pages making it difficult for me to see what buttons to press

Scenario Title: Elderly mode during Device rotation

Given Elderly mode is active, and I prefer to have my phone sideways

When I rotate my phone sideways

Then the text and buttons keep their position and keep their larger sizes maintaining a simple layout

Story ID: 13

Feature Title: Student discounts

Story Title: Student tariff pricing

Story Narrative: As a student who commutes to university, I want to be able to travel affordably. Having the opportunity to get a discounted travel with student pricing would release a lot of stress.

Story Size: Medium

Acceptance Criteria Scenarios:

Scenario Title: Student tariff pricing is displayed

Given My student ID has been verified

When I go to see the price for my journey

Then the displayed student price is displayed

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I can complete the purchase without any issues.

Scenario Title: Unverified student ID

Given My student ID has been expired or not verified

When I select my journey to see prices

Then The app shows an error displaying discount not available

App displays an error message

Gives user reason why it's not working

User can either pay full price or resolve the issue

Scenario Title: My student ID was previously verified and now expires.

Given My student ID expires a month after verification

When I try to purchase a student ticket

Then The app displays a discounted rate

I can either purchase discounted rate as the system has not unverified my ID

Story ID: 17

Feature Title: Online Bus/Train Timetabling

Story Title: Viewing and interacting with an update-to date real-time bus/train timetable

Story Narrative: I am a daily commuter that relies on real-time timetable information for buses and trains, so can play efficiently, and avoid unexpected travel disruptions

Story Size: Large

Acceptance Criteria Scenarios:

Scenario Title: Daily commuter successfully views an update-to-date timetable

Given I am a Daily commuter travelling frequently in Dublin and I want to plan my trip to my relative's house in advance

When I try to access the bus/train timetable a few days before my journey

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Then I see an accurate up – to-date timetable with information on all the trains and buses allowing me to get the earliest bus to my destination after I arrived

Scenario Title: Daily commuter stranded due to out-of-date information

Given I commute to work daily

And rely on buses to travel to work

When the bus is cancelled due to severe weather problems

Then the timetable was not updated effectively to reflect this cancellation leaving me stranded at the bus stop after work waiting, the timetable should have updated immediately to notify users of cancellations.

Scenario Title: Daily commuter, mislead to buy a ticket for the wrong date

Given I commute daily via buses

And rely on buses to do my shopping, I accessed the buy page to buy my ticket

When the page defaulted to the next day

Then I mistakenly bought a ticket thinking I could use it today, causing a lot of inconvenience, the system should have clearly displayed the selected date before confirming purchase

Story ID: 21

Feature Title: In app ticket purchase

Story Title: Purchase ticket

Story Narrative: As a commuter I want to be able to purchase my ticket easily and reliably, highlighting any issue immediately so that I can get to my volunteering work on time.

Story Size: Large

Acceptance Criteria Scenarios:

Scenario Title: Purchase tickets successfully

Given I have created an account, and logged in

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I am at the section to purchase a ticket

When I select my departure station and my arrival station, a list of fares will appear and I purchase one.

Then The ticket is automatically connected onto my account

I can view the ticket at any time until it's no longer valid.

Scenario Title: Unable to validate the ticket as I have no internet

Given I purchase my ticket earlier on in the day

And I'm at the station ready to use my ticket

When I go to validate my ticket but have no mobile data

Then I cannot access my ticket to validate

I'm unable to use the ticket for travel

Scenario Title: Duplicate ticket

Given I already purchase a monthly ticket on the app yesterday

I forgot that I purchased this ticket

When I try to purchase another monthly ticket for the same date

Then The app throws me an error message

The reasoning is there is another ticket the exact same already purchased

But it's took payment for both tickets

I then must get in contact about refunding one of the tickets

Story ID: 27

Feature Title: Currency conversion

Story Title: Converts ticket prices into a different currency

Story Narrative: As a person who is sightseeing in Dublin, I want to convert ticket prices within the app. To help me keep track of costs, so I stay within the budget and can enjoy my time travelling.

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Story Size: Medium

Acceptance Criteria Scenarios:

Scenario Title: Ticket price converts successfully

Given I am in Dublin

And have access to the internet to use the app

When I convert the price of the ticket to USD dollars

Then the app displays the amount converted

I can decide to purchase ticket or not

Scenario Title: Exchange rate goes down during conversion

Given I'm on the app about to purchase a ticket

I try to convert my ticket price into USD

When The exchange rate service has become unavailable

Then The app shows an error message

The converted price doesn't show up

I still feel frustrated that I cannot see the price and stress whether I can afford it

Scenario Title: Currency isn't supported by the API

Given I have just arrived in Dublin, but want to find an affordable way to travel to Dublin

I try to convert ticket prices into my home currency

When The app doesn't have my home currency as the API server does down

Then the top up card should display the new value of money on the card.

The app displays multiple error messages

The conversion feature is temporarily unavailable

I cannot see how much of my budget is being spent on travel, which is causing me a lot of stress.

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Spike Stories

Spike Story 1

Spike Title: Currency Converter

Spike Type: Functional

Spike Narrative: To stay on budget our tourists and citizens needs to be able to convert ticket prices to their own currency.

Spike Story Description:

Failure of the Currency Converter function causes a trouble for buying any kind of ticket in the travel app, unavailability to make purchase through non-euro accepting bank accounts. A lot of tourists may not be able to travel and will crowd streets trying to reach their destination or find the way to buy physical ticket, withdraw their money in ATM or trade concurrency.

Spike Story 2

Spike Title: Elderly Mode UI Consistency

Spike Type: Functional

Spike Narrative: For our older users to use our app, we need to make it more user friendly for them. By implying large text, buttons and icons.

Spike Story Description:

To add Elderly Mode into our app, we need to ensure compatibility with various devices and sizes. If we added large text and buttons to suit our screen size, the user screen size will be different therefore will not suit. We can manage this by adding more coding into our interface, and we can test this on different users before releasing to ensure it works fully. The differences within front scaling, spacing and padding can create confusing for users leaving them less accessible to our app. This spike can only be proofed through testing by using our many user interfaces prototypes, plus user feedback from a selected number of users. This all ensures our app will allow elderly mode to work correctly, by providing more independence to those who require it.

Spike Story 3

Spike Title: Purchase tickets online and offline

Spike Type: Technical

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Spike Narrative: For customers to travel without any hassle, we can have an app that allows users to purchase tickets through the app. Which can also be used in the event of no internet connection.

Spike Story Description: This spike allows us to investigate the technical risks that come with this feature, such as security, data synchronisation and device compatibility. For tickets to be available offline they need to be stored locally. However, this can lead to sensitive data being leaked. iPhone and android may have different ways of storing data. Ensure users can view tickets offline such as airplane mode.

Spike Story 4

Spike Title: Real-Time Timetable Data Reliability

Spike Type: Technical

Spike Narrative: To provide accurate journey planning, the app needs to reliably retrieve up-to-date data and ensure data Integration

Spike Story Description:

This spike caters for the investigation of reliable journey planning through multiple sources of data from transport services and ensuring that the data is correctly synchronized and updated within the system to provide real-time information to the user. This spike will explore how frequently updates occur, ensuring consistent data across the system as well as identifying potential challenges in the aspect of integrating the third-party information effectively. Attention will be given to edge cases such as disruptions to the service to ensure our system is robust. The goal is to discover how well the system architecture supports accurate journey planning under normal and exception conditions.

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Story Map

Story points – (Small = 1-3) (Medium = 4-6) (Large = 7-10)

Features	Elderly Mode	Accessibility - Audio/Visual	Currency converter	Accessibility Friendly routes	Student Tariff Pricing	Online Bus/Train Timetables	In-App Ticket Purchases	Multi-Language support
Sprint 1 Goals of this sprint-Create the core skeleton of the app Capacity-44/50	Spike – To give elderly people the opportunity, a team will create a function that sizes all the text, icons and other UI elements. (3)	As a visually impaired user I want a high contrast colour scheme, so icons and texts are easier to see (5) TD[5]	As a person who is sightseeing in Dublin, I want to convert ticket prices within the app. To help me keep track of costs, so I stay within the budget and can enjoy my time travelling. (6) TD [19]	Spike- In-Order to provide Wheelchair friendly routes, we need to investigate data sources/ APIS (3)	As a student who commutes to university, I want to be able to travel affordably. Having the opportunity to get a discounted travel with student pricing would release a lot of stress. (5) TD [8,9,10,11]	As a commuter I want to see real time bus departures for chosen stops so I can time my journey more accurately (8) TD [3,6,7,8,10,12,13,14]	Spike- To sell tickets in the app, the team needs to research compliant payment providers (3)	

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	As an elderly person I would like a user interface that has large text, big buttons and clear icons (3) TD [5]						As a commuter I want to be able to purchase my ticket easily and reliably, highlighting any issue immediately so that I can get to my volunteering work on time. (8) TD [1,15,16,17,19]	
Sprint 2 Goals of this sprint – Accessibility and inclusion Capacity-33/50		As a blind user, I would need the app to have audio assistance so I can book. (7) TD [1]	As an international tourist I would like the tickets converted into my home currency so I can prepare my transport (6) TD [19,20,21]	As a wheelchair user I want journey options that will include wheelchair friendly stations (8) TD [6]		As an ordinary app user, I want to have access to bus /train timetable anywhere at any time so I could plan my route (5) TD [3,6,7,12]	As an international tourist, I want to be able to view information in my preferred language so I can book correctly (7) TD [18]	

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Sprint 3 Goals for this sprint- User personalisation Capacity- 41/50		As a disabled user, I want a simple start/ stop listening control for voice commands (5) TD [2]	As a user, I want it to use my location and default to the currency in my location (6) TD [20]		As a student I would like a verified student account so that I can access discounts for my travel instantly without needing to upload proof every time (7) TD [9]	As a daily commuter whose phone has poor battery, I would like the option to print the online timetable without needing to worry about my phone battery (3) TD [13]	As a user I want to securely store a payment method so I can buy tickets quickly (7) TD [15]	As an international tourist, I want the app to detect my country and automatically put it in my language for me (6) TD [18]
		As a user I want to switch between light and dark themes to help with eye reducing strain (2)					As a frequent Traveller, I want to top up my travel card to avoid queues (5) TD [16]	

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Sprint 4		As a disabled user I would like to search the app using only my voice (8) TD [2,3,4]	Spike- To provide a currency converter we need to investigate APIS in exchange rate (4)	As a disabled user I want to see an option for step free routes so I can get around easily (8) TD [7]	As a student I want tickets to automatically display my discounted price (4) TD [10]			Spike- to support multiple languages, the team need to select a framework and define a translation method (5)
Goals for this sprint- Advanced accessibility / quality of life improvement								
Capacity 33/50								
		As a daily user, I want my contrast and font option saved so I don't need to keep enabling this (4) TD [5]						
Sprint 5	As an elderly person I want clear confirmation of my	As a daily user, I would like to be able to book using my voice to	As a user, I want it to show me the exchange rate so I can be		As a student on a tight budget, I want to see the	As a daily commuter, I would like it to automatically recommend my	As a student on a budget, I would like my in-app purchases sent	

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Goals for this sprint- Increased quality of life	purchases and any errors that may occur (4) TD [1]	save time typing (6) TD [1,2,4]	more informed about the costs (5) TD [21]		percentage and the amount of the discount (3) TD [11]	most common route, to save time entering details again (7) TD [14]	to my email for 2 nd way of confirmation (3) TD [17]	
Capacity-33/50		As a user, I would like the system to repeat if it doesn't understand my commands (5) TD [4]						

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Development Tasks

Feature: Accessibility – Audio/Visual

User Story ID: S3

User Story Title: As a visually impaired individual, I would need the app to have audio assistance so I can book tickets

Task ID: T1

Task Title: Incorporate A text-to Speech Search Engine

Task Size: Large

Dependencies:

- None

Subtasks:

- *Incorporate a text to speech engine in all interactive elements of the application*
- *Incorporate a speech to text engine with input handling.*
- *Implement audio navigation*
- *Make sure all UI components are compatible with WCAG (web content accessibility guidelines)*
- *Create an interface for audio assistance.*
- *Apply error handling for audio feedback*
- *Conduct testing to ensure this works.*

Task ID: T2

Task Title: Audio feedback form

Task Size: Medium

Dependencies:

- T1

Subtasks:

- *Create an accessible form*
- *Apply rating and evaluation components to this form.*
- *Incorporate some feedback fields.*

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- Connect the form to the backend service to have a strong audio feedback submissions in the systems database.
- Add audio guidance into the form,
- Add validation and error handling into the form
- Conduct testing to ensure everything works as expected

Task ID: *T3*

Task Title: *Incorporate voice command integration*

Task Size: *Medium*

Dependencies:

- *T1*

Subtasks:

- Incorporate speech to text engine, to process input and return it as text.
- Create a voice recognition system, so it can read the components to the user.
- Develop voice navigation controls for the user to move throughout the app
- Develop text to speech feedback, whenever the user clicks a component, or an error happens.
- Add logic so if there is background noise and the voice isn't picked up, make sure there is a fallback system.
- Let visually impaired use their voice to enter data, into input fields for easier use of app.
- Perform testing on different operating systems, devices, platforms and in different areas with variety of noise levels, to make sure the voice input works correctly and is reliable.

Task ID: *T4*

Task Title: *Accessibility for all user interfaces*

Task Size: *Medium*

Dependencies:

- *T1*
- *T2*
- *T3*

Subtasks:

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- *Conduct testing every component of the user interface, to ensure it complies with screen readers. Ensuring it all is compliant with legislation such as WCAG.*
- *Tidy up the UI layouts so users can clearly direct themselves around the application in a logical order by either using an assistive device or a keyboard.*
- *Improve the interactive components on the UI interface, so it includes a title and description. This helps the users' accessible devices read the text out loud to understand what each component does.*
- *Develop logic so when a notification pops up, it doesn't change the user's focus. As navigation can be hard when visually impaired, the logic can help keep it simplified and easy for the user.*
- *Perform testing on this system to make sure everything works as it should, allowing this to be more accessible for those impaired.*

Feature: Student Pricing for University commuters

User Story ID: S13

User Story Title: As a university student who commutes, I want to be able to travel affordably.

Task ID: T5

Task Title: Incorporate Student pricing system

Task Size: Large

Dependencies:

- None

Subtasks:

- *Create logic so a student can verify their status; by verifying their student email address to make sure they are eligible for the student tariff.*
- *Develop an algorithm so when student status is verified, it automatically calculates and applies the discounted price.*
- *Make sure the algorithm is incorporated into the system, so the student can pay and view the ticket without any issues.*
- *Perform testing with many student profiles with verification, to see if it works if not to make sure there is error handling.*
- *Allow students to upload their student ID's securely in compliance with GDPR regulations.*
- *Include a section where students can input data of their Student ID in the account settings, such as expiry dates.*

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Task ID: T6

Task Title: *Display the discounted prices in the booking system*

Task Size: *Medium*

Dependencies:

- T5

Subtasks:

- *When student status has been verified, create logic so the system picks up on this.*
- *Create an algorithm for discounted prices.*
- *Make sure the API service supports this, and it complies with student prices.*
- *Add more fields into the database to securely hold user's student details.*
- *Change the UI to be able to fit all the prices*
- *Testing this system to ensure that it works*
- *Test the system so if the user is not a student, it does not show the student prices*

Task ID: T7

Task Title: *Use the university API to verify student status*

Task Size: *Large*

Dependencies:

- T5

Subtasks:

- *Look at the APIs for university's*
- *Define what parameters, authentication and handling is required*
- *Create a system that calls the university's API*
- *Make sure the system can handle this API*
- *Create a user interface for the university's API*
- *Implement error handling incase the APIs are down, or the student is not found*
- *Make sure all laws and legislations are in place*
- *Test this system to ensure it works with many cases*

Task ID: T8

Task Title: *Test student discounts across different platforms ensuring functionality.*

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Task Size: *Medium*

Dependencies:

- *T5*
- *T6*

Subtasks:

- *Start creating testing plans for each operating system, to ensure functionality*
- *Create different sceneries for every operating system.*
- *Conduct testing ensuring everything is fully operational as well as all data is showed on user interfaces*
- *If any errors display, develop more code to work around these errors*
- *Conduct testing again to ensure everything is fully functional.*
- *Test positive and negative scenarios and ensure the system handles it well.*
- *Document all testing results and conduct this testing regularly.*

Feature: Online Bus/timetable

User Story ID: S17

User Story Title: As a commuter I want to see real time bus /train departures for chosen stops so I can time my journey more accurately

Task ID: *T9*

Task Title: Implement Real-Time bus/train departure API integration

Task Size: *large*

Dependencies:

- *None*

Subtasks:

- *Define what is needed for an API system, and the constraints of this.*
- *Create the API system to retrieve the live data*
- *Implement logic so if the API fails, it shows last known data*
- *Incorporate this into the application*
- *Update the UI suit this*
- *Test this application with this real time data*
- *Monitor the system for any bugs, or updates needed.*

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Task ID: T10

Task Title: As an ordinary app user, I want to have access to bus /train timetable anywhere at any time so I could plan my route

Task Size: Medium

Dependencies:

- T9

Subtasks:

- *Update the timetables so when downloaded it makes sure it has all required data on it*
- *Secure the timetable onto the API service*
- *Users must allow the app to use local storage, if not error message needs to present*
- *Make sure that anyone can download a timetable by ensuring its cross-platform compatibility and compliance with WCAG.*
- *Add logic so users can customize a timetable. E.g. travelling from Connolly to the Grand Canal Dock rather than the entire timetable.*
- *Test this to make sure it works and has a clear layout when downloaded.*

Task ID: T11

Task Title: As a daily commuter whose phone has poor battery, I would like the option to print the online timetable without needing to worry about my phone battery

Task Size: Medium

Dependencies:

- T9
- T10

Subtasks:

- *Update the timetables so when user wants to print a timetable, there's a button indicating they can.*
- *Secure the timetable onto the API service*
- *Users must connect to a local printer; if not successful, an error message needs to be presented.*
- *Test to make sure that anyone can download a timetable off any device.*
- *Add logic so users can customize a timetable. E.g. travelling from Connolly to the Grand Canal Dock rather than the entire timetable.*

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- *Test this to make sure it works and has a clear layout when printed out.*

Task ID: *T12*

Task Title: As a daily commuter, I would like it to automatically recommend my most common route, to save time entering details again

Task Size: *Medium*

Dependencies:

- *T9*
- *T10*
- *T11*

Subtasks:

- Define what criteria will be needed for this to work successfully.
- User will have to allow the app to collect their data
- Build a system that will keep track of user's routes and calculate which one is used the most. If none exists it will recommend popular routes near you.
- Develop your UI to suit this new system. For example, adding another component under the navigational panel.
- Create a field in settings which allow the user to clear their data at any time.
- Complete testing on this system with different scenarios.
- Imply logic to ensure the app is optimising, when possible, for better performance.

Test Cases

Test ID: TC1	Test Description: <i>Verification that elderly mode enlarges all elements across each page on the app</i>		
Story ID: S1	Story Narrative: As an elderly user, I want to be able to enable an Elderly Mode so that I can easily read text and navigate using larger buttons and icons due to accessibility needs		
Step ID	Step Description	Test Data	Expected Result

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1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigating to profile	Click User Profile Icon	Drop down appears with list of options
3	Selecting settings	Click Settings through drop down option	User is directed to settings page
4	Enabling Elderly Mode	Click the Elder mode Toggle and select confirm button	App enables Elderly mode
5	App refresh	Automatic refresh of App occurs after confirmation is selected	App UI reloads with larger elements (buttons and Icon/images)
6	Navigate to Home page	App logo is clicked	User is directed to Home page with enlarged UI elements
7	Navigate to Timetable page	Timetable page button is clicked	User is directed to Timetable page with enlarged UI elements
8	Navigate to contact page	Contact page button is clicked	User is directed to Contact us page with enlarged UI elements

Test ID: TC2	Test Description: <i>Checking that Elderly Mode fails to apply consistent sizes of elements across certain pages</i>		
Story ID: S1	Story Narrative: As an elderly user, I want to be able to enable an Elderly Mode so that I can easily read text and navigate using larger buttons and icons due to accessibility needs		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account

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2	Navigating to profile	Click User Profile Icon	Drop down appears with list of options
3	Selecting settings	Click Settings through drop down option	User is directed to settings page
4	Enabling Elderly Mode	Click the Elder mode Toggle and select confirm button	App enables Elderly mode
5	App refresh	Automatic refresh of App occurs after confirmation is selected	App UI reloads with larger elements (buttons and Icon/images)
6	Navigate to Home page	App logo is clicked	User is directed to Home page with properly enlarged UI elements
7	Navigate to Timetable page	Timetable page button is clicked	Issue: User is directed to Timetable page with enlarged UI elements Elements are enlarged only slightly and are smaller than the elements on the home page.
8	Navigate to contact page	Contact page button is clicked	Issue: User is directed to Contact us page with enlarged UI elements Text above the user input fields is not enlarged

Test ID: TC3	Test Description: <i>Checking that Elderly Mode keeps layout consistent and elements enlarged on device rotation</i>
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Story ID: S1	Story Narrative: As an elderly user, I want to be able to enable an Elderly Mode so that I can easily read text and navigate using larger buttons and icons due to accessibility needs		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigating to profile	Click User Profile Icon	Drop down appears with list of options
3	Selecting settings	Click Settings through drop down option	User is directed to settings page
4	Enabling Elderly Mode	Click the Elder mode Toggle and select confirm button	App enables Elderly mode
5	App refresh	Automatic refresh of App occurs after confirmation is selected	App UI reloads with larger elements (buttons and Icon/images)
6	Navigate to Home page	App logo is clicked	User is directed to Home page with properly enlarged UI elements
7	Device rotation to landscape	Device physically rotated to landscape by user	Layout of elements is correctly positioned and resembles the original layout with enlarged UI elements kept enlarged
8	Device rotation to portrait	Device physically rotated to portrait by user	Layout and UI elements return to match default home page before landscape rotation
9	Navigate to Timetable page	Timetable page button is clicked	Issue: User is directed to Timetable page with enlarged UI elements

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			Elements are enlarged only slightly and are smaller than the elements on the home page.
10	Device rotation	Device physically rotated to landscape by user	Layout of elements is correctly positioned and resembles the original layout with enlarged UI elements kept enlarged
11	Device rotation	Device physically rotated to portrait by user	Layout and UI elements return to match default timetable page before landscape rotation
12	Navigate to contact page	Contact page button is clicked	Issue: User is directed to Contact us page with enlarged UI elements Text above the user input fields is not enlarged
13	Device rotation to landscape	Device physically rotated to landscape by user	Layout of elements is correctly positioned and resembles the original layout with enlarged UI elements kept enlarged
14	Device rotation portrait	Device physically rotated to portrait by user	Layout and UI elements return to match default contact page before landscape rotation

Test ID: TC4	Test Description: Verification that student ID can be verified and displays discounted rate upon viewing, and deducts correct amount based on new total
Story ID: S13	Story Narrative: As a student who commutes to university, I want to be able to travel affordably. Having the opportunity to get a discounted travel with student pricing would relieve a lot of stress.

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Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigating to profile	Click User Profile Icon	Drop down appears with list of options
3	Selecting Student	Click Student through drop down option	User is directed to Student page
4	Verify Student ID	Correct Student ID number is entered and verified by the system	Refreshes and Informs user that student ID has be verified
5	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page
6	Search transport journey	User enters route: "Dublin" to "Belfast"	Displays all the journeys and discounted price besides
7	Journey purchased	Selects 9.am Train and enter payment details	Ticket successfully purchased and User is charged based on the student discount

Test ID: TC5	Test Description: Check to see if expired student ID doesn't allow discounted price and presents error message to user		
Story ID: S13	Story Narrative: As a student who commutes to university, I want to be able to travel affordably. Having the opportunity to get a discounted travel with student pricing would release a lot of stress.		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigating to profile	Click User Profile Icon	Drop down appears with list of options

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3	Selecting Student	Click Student through drop down option	User is directed to Student page
4	Verify Student ID	Correct Student ID number is entered, which is past its expiry date	Verification fails and error message is presented to user showing not verified
5	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page
6	Search transport journey	User enters route: "Dublin" to "Belfast"	Issue: Displays all the journeys and discounted price is not shown, with only full price shown
7	Purchasing Ticket	Pay full ticket or contact University for new ID	Students can successfully purchase tickets at full price or retry verification if they get renewed ID with up-to date expiry date

Test ID: TC6	Test Description: Check to see if previously verified ID, that then expires blocks discount rate and removes the ID		
Story ID: S13	Story Narrative: As a student who commutes to university, I want to be able to travel affordably. Having the opportunity to get a discounted travel with student pricing would release a lot of stress.		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigating to profile	Click User Profile Icon	Drop down appears with list of options
3	Selecting Student	Click Student through drop down option	User is directed to Student page

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4	Check student ID status	Previously verified ID is now expired	Issue: Displays that user ID is still verified, this should now say Unverified
5	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page
6	Search transport journey	User enters route: "Dublin" to "Belfast"	Issue: Displays all the journeys and discounted prices are shown even though student ID has now expired. This should show only show full price
7	Purchasing Ticket	Pay full ticket or contact University for new ID	Issue: Students can successfully purchase tickets at a discounted price. This should have charged full price

Test ID: TC7	Test Description: Verification that a user can view real-time information on the timetable for buses and trains		
Story ID: S17	Story Narrative: I am a daily commuter that relies on real-time timetable information for buses and trains, so I can plan efficiently, and avoid unexpected travel disruptions		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page

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3	App orders the modes of transport based on earliest departure	Upon entering page	Journeys should be ordered from earliest departure to latest
4	Search transport journey	User enters route: "Dublin" to "Cork"	Refreshes and displays all the journeys with available modes of transport
5	Filters For specific Day	User enters a filter for the date 17/10/2025	Updates timetable and show the modes of transport for this day with accurate arrival date and duration times for these transports
6	Filter for specific mode of transport	User enters a filter to show only buses	Timetable updates and shows only Bus related information for the 17/10/2025

Test ID: TC8	Test Description: Verification timetable will update in event of transport cancellations to show accurate information		
Story ID: S17	Story Narrative: I am a daily commuter that relies on real-time timetable information for buses and trains, so can plan efficiently, and avoid unexpected travel disruptions		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page
3	App orders the modes of transport based on earliest departure	Upon entering page	Journeys should be ordered from earliest departure to latest
4	Search transport journey	User enters route: "Dublin" to "Belfast"	Refreshes and displays all the journeys with available modes of transport

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5	Bus cancellation	Weather conditions means that buses from Dublin to Belfast are cancelled	Issue: Timetable is not yet updated to reflect this
6	Filters for current day	User enters a filter for the date 15/10/2025	Issue: App displays all modes of transport for this day as available
7	Filter for specific mode of transport	User enters a filter to show only buses	Issue: Timetable updates and shows Bus related information for the 15/10/2025 as available should show cancelled buses as cancelled.

Test ID: TC 9	Test Description: Verification that ticket purchasing page clearly displays the date picked on timetable page		
Story ID: S17	Story Narrative: I am a daily commuter that relies on real-time timetable information for buses and trains, so can plan efficiently, and avoid unexpected travel disruptions		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page
3	App orders the modes of transport based on earliest departure	Upon entering page	Journeys should be ordered from earliest departure to latest
4	Search transport journey	User enters route: "Dublin" to "Cork"	Refreshes and displays all the journeys with available modes of transport

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5	Filters For specific Day	User enters a filter for the current date 14/10/2025	Updates timetable and show the modes of transport for this day with accurate arrival date and duration times for these transports
6	Filter for specific mode of transport	User enters a filter to show only buses	Timetable updates and shows only Bus related information for the 14/10/2025
7	Select ticket	User selects the 10am Bus for current date which is 14/10/2025 and clicks purchase	Issue: When directed to the payment page, and the ticket details, it automatically defaults the ticket to the next day (15/10/2025) as it didn't pull through the date, they selected on the previous page which the app should have.

Test ID: TC 10	Test Description: Verifying that a user can purchase a ticket from their app's balance and view the ticket until it expires		
Story ID: S21	Story Narrative: As a commuter I want to be able to purchase my ticket easily and reliably, highlighting any issue immediately so that I can get to my volunteering work on time.		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page

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3	App orders the modes of transport based on earliest departure	Upon entering page	Journeys should be ordered from earliest departure to latest
4	Search transport journey	User enters route: "Dublin" to "Belfast"	Refreshes and displays all the journeys with available modes of transport
5	Filters For specific Day	User enters a filter for the current date 20/10/2025	Updates timetable and show the modes of transport for this day with accurate arrival date and duration times for these transports
6	Filter for specific mode of transport	User enters a filter to show only buses	Timetable updates and shows only Bus related information for the 20/10/2025
7	Purchase ticket	User selects a journey and clicks purchase	Directly deducts the amount from the balance in their app, if not enough, highlighting this in a red banner at the top of app
8	View ticket	User clicks on My tickets under profile Icon	Displays the ticket QR code and timer expiry date (end of day)
9	QR code scanned	User Scans QR code at bus	Displays on scanner as verified indicating ticket was bought and registered successfully through the app

Test ID: TC 11	Test Description: Checking that a user can purchase a ticket, but it can't be validated by scanner offline
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Story ID: S21	Story Narrative: As a commuter I want to be able to purchase my ticket easily and reliably, highlighting any issue immediately so that I can get to my volunteering work on time.		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page
3	Purchase ticket	User selects a journey and clicks purchase	Directly deducts the amount from the balance in their app, if not enough, highlighting this in a red banner at the top of app
4	View ticket	User clicks on My tickets under profile Icon	Displays the ticket QR code and timer expiry date (end of day)
5	Arrive at transport station	User tries to access QR code offline	Issue: The QR code does not appear, making it impossible to validate at the station. It should've displayed an error message and stored the QR code for offline use

Test ID: TC 12	Test Description: Checking that a user is prevented from purchasing a duplicate ticket for the same day and highlights issue immediately		
Story ID: S21	Story Narrative: As a commuter I want to be able to purchase my ticket easily and reliably, highlighting any issue immediately so that I can get to my volunteering work on time.		
Step ID	Step Description	Test Data	Expected Result

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1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page
3	Purchase ticket	User selects a journey and clicks purchase for monthly pass	Directly deducts the amount from the balance in their app, if not enough, highlighting this in a red banner at the top of app
4	View ticket	Click profile icon and select my tickets	Show that the ticket is there successfully
5	Purchasing duplicated ticket	User selects same journey and clicks purchase for the next day	Issue: App allows users to proceed to enter details, and click purchase, throws an error message afterwards highlighting duplicate purchase
6	View Travel card balance	User clicks profile icon, and Travel card	Issue: Shows that the duplicated ticket has withdrawn from the balance, which it should have denied

Test ID: TC 13	Test Description: Verifying that the price of tickets is successfully converted into pounds		
Story ID: S27	Story Narrative: As a person who is sightseeing in Dublin, I want to convert ticket prices within the app. To help me keep track of costs, so I stay within the budget and can enjoy my time travelling.		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account

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2	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page showing all the journeys and their prices in Euros
3	View prices	User selects a journey and clicks on a journey ticket which is 10 euro	Directs user to payment page
4	Convert currency	User clicks currency dropdown and selects pound	Converts 10 euro to pounds considering exchange rate (8.76) successfully and displays to user
5	Purchasing ticket	User clicks enter payment information and clicks purchase	Message stating ticket purchase successful

Test ID: TC 14	Test Description: Checking that the app displays an error when exchange rate service is unavailable		
Story ID: S27	Story Narrative: As a person who is sightseeing in Dublin, I want to convert ticket prices within the app. To help me keep track of costs, so I stay within the budget and can enjoy my time travelling.		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page showing all the journeys and their prices in Euros
3	View prices	User selects a journey and clicks on a journey ticket which is 10 euro	Directs user to payment page
4	Convert currency	User clicks currency dropdown and selects pound	Issue:

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			App can't convert euro to pounds as the exchange rate API went down making this service unavailable and not displaying anything, app should have displayed an error message letting user know why
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Test ID: TC 15	Test Description: Checking that the app supports all currencies		
Story ID: S27	Story Narrative: As a person who is sightseeing in Dublin, I want to convert ticket prices within the app. To help me keep track of costs, so I stay within the budget and can enjoy my time travelling.		
Step ID	Step Description	Test Data	Expected Result
1	<i>Open and log in to the app</i>	Valid username and Password are entered	App successfully loads and user logs in to account
2	Navigate to Timetable page	Timetable page button is clicked	Directs user to timetable page showing all the journeys and their prices in Euros
3	View prices	User selects a journey and clicks on a journey ticket which is 10 euro	Directs user to payment page
4	Convert currency	User clicks currency dropdown looking for Chinese currency Yuan	Issue: The dropdown menu doesn't appear to have Yuan as this is not supported by API, meaning user can't select and make a payment due to it not being a supported currency.